

Building Bigger Shapes from Smaller Ones

counting the small squares in a composed figure, naming the new shape, comparing two builds,
and finishing a rectangle

Directions

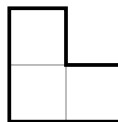
- Small shapes can be fitted together to build a bigger shape. The pieces must not overlap and must leave no gaps.
 - The thin lines show the small squares. The thick line shows the outside of the built shape.
 - Touch each small square once as you count, so none is counted twice.
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1. How many small squares cover this shape?



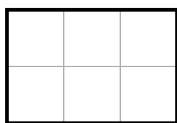
Number of small squares: _____

2. How many small squares cover this shape?



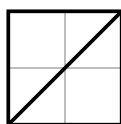
Number of small squares: _____

3. How many small squares cover this rectangle?



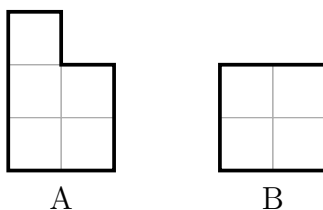
Number of small squares: _____

4. Two matching triangles are joined along their longest edges. The thick line shows where they meet.



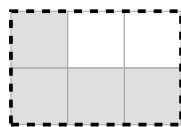
- (a) Name the new shape the two triangles make. _____
- (b) How many small squares does the new shape cover? _____

5. Two shapes were built from small squares.



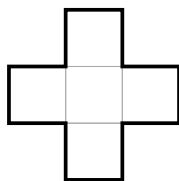
Which shape uses more small squares? Write $<$, $>$, or $=$: A _____ B

6. The dashed line shows the rectangle to be built. The shaded squares are already in place.



How many more small squares are needed to fill the rectangle? _____

7. How many small squares cover this shape?



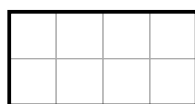
Number of small squares: _____

8. Two matching corner pieces are fitted together. The thick line inside shows where they meet.

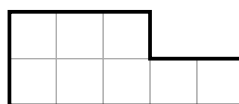


- (a) Name the new shape the two pieces make. _____
- (b) How many small squares does the new shape cover? _____

9. Two shapes were built from small squares. They do not look alike.



A



B

- (a) Count the small squares in each shape. Write $<$, $>$, or $=$: A _____ B
- (b) Explain how two shapes that look so different can use the same number of small squares.

10. The dashed line shows the rectangle to be built. The shaded squares are already in place.



- (a) How many more small squares are needed to fill the rectangle? _____
- (b) Explain how you worked that out without placing the squares one at a time.
